

ISEPT: Image-Based Selection and Execution Framework for Pair Trading

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Abstract

Pair trading is a statistical arbitrage strategy that seeks market-neutral profits by hedging two assets. Most earlier studies separate pair selection and execution, running each step in its own model. This split prevents trading results from informing selection and often overfits a small set of assets. In addition, many approaches rely on raw or aggregated prices, which struggle to capture the visual patterns and cross-asset interactions that influence returns. Accordingly, we introduce ISEPT, an end-to-end framework that works directly with candlestick chart images and joins pair selection with trading. A convolutional autoencoder (CAE) converts monthly candlestick images into stock-level latent vectors. For each pair, the concatenated vectors pass to a multilayer perceptron (MLP) that predicts the next month's Sharpe ratio. At month-end, realized trading results feed back into the MLP as fresh training data, letting the model adjust to current market conditions and refine pair rankings continuously. Tests on a long historical span of daily Open-High-Low-Close (OHLC) data for S&P 500 constituents show that ISEPT outperforms traditional approaches. Both return on investment and Sharpe ratio improve markedly, indicating that the framework delivers steady long-term gains.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → **Neural networks**.

Keywords

Pair Trading, Statistical Arbitrage, Candlestick Image Representation, Convolutional Autoencoder (CAE), Sharpe Ratio

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1 Introduction

Pair trading, a statistical arbitrage technique that exploits temporary deviations between the prices of two highly correlated assets.

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Since its introduction in the late 1980s [11], its use in both practice and academia has grown steadily. By swiftly correcting mispricings, it improves market efficiency and has delivered long-run profits across a range of asset classes, including equities, futures, and cryptocurrencies [7].

Most previous studies adopt a two-step process for pair trading: pair selection followed by trade execution. For pair selection, traditional statistical metrics such as cointegration tests or distance measures are used to identify candidate pairs, and fixed threshold rules then guide execution [8, 43]. Although some studies apply machine learning or deep learning models to predict spreads or generate signals, they still keep pair selection separate from execution, leaving the two stages disconnected [10, 16]. Consequently, researchers have observed that selected pairs may yield only small spreads or that trading performance deteriorates because the models overfit the short list of chosen assets [1]. Furthermore, by feeding raw numerical time-series data straight into models, most earlier work overlooks the visual patterns in candlestick charts, patterns that traders have relied on for decades [36].

To address these issues, we propose an Image-Based Selection and Execution Framework for Pair Trading (**ISEPT**), an end-to-end framework that (i) jointly optimizes pair selection and trading and (ii) learns visual cues through image representations. We encode cross-asset relations by turning price series into candlestick images and close the feedback loop by using realized trading results as training data for next-month pair selection [15, 18]. The framework first extracts latent vectors from monthly candlestick images with a convolutional autoencoder (CAE). It then concatenates the two stock vectors and feeds the result to a multilayer perceptron (MLP) that estimates the Sharpe ratio for the following month. Standard trading rules govern position entry and exit [11, 43]. At each month-end, the realized trading result becomes the label for updating the MLP, letting the model refresh its selection rules continually. As a result, empirical tests on daily Open-High-Low-Close (OHLC) data for S&P 500 constituents from January 2004 through June 2024, roughly twenty years, show that ISEPT outperforms traditional statistical approaches in cumulative return and Sharpe ratio, supporting the usefulness of the image-based representation.

The key highlights of the proposed model are:

- (1) New representation for pair trading captures both within-asset and cross-asset patterns by converting OHLC data into candlestick images.
- (2) Unified framework retrains monthly using realized trading results, keeping pair-selection rules adaptive.
- (3) Long-horizon (nearly twenty years) empirical analysis shows the proposed approach outperforms standard statistical baselines in real-world pair trading.

2 Related Work

2.1 Classical Pairs Trading

Distance-based methods, especially the Euclidean sum of squared differences (SSD) [11, 29], were among the first techniques for pairing assets by comparing their price paths. Gatev et al. studied U.S. equities from 1962 to 2002, selected the pairs with the smallest SSD, and found that the resulting spreads moved so narrowly that almost no trading opportunities emerged [11, 46]. A later meta-review that catalogued the theoretical shortcomings of this approach reached the same conclusion [21, 31].

To remedy such shortcomings, the Engle-Granger cointegration test became widespread: if the residuals are stationary, the price difference is assumed to be mean-reverting, and the pair is considered tradable. However, many studies note that past cointegration often fails to persist [5]. Copula-based criteria have been proposed to handle non-normal spread distributions, and wavelet transforms to remove common co-movement factors [6, 9]. Despite these advances, the methods still depend solely on historical statistics, so their performance is fragile when market conditions shift.

2.2 Artificial Intelligence for Pair Trading

Recent research has turned to AI methods to overcome the limits of traditional statistics and to optimize pair selection and trading together. Most machine learning and deep learning models are applied at the execution stage, where they predict spread direction or adjust entry and exit thresholds. For example, Flori et al. used Long Short-Term Memory (LSTM)-based probabilities of spread widening and narrowing to schedule trades and achieved higher returns than standard statistical indicators [10]. Kuo et al. proposed a two-stage deep model that updates entry and exit thresholds dynamically and filters out unprofitable trades, improving overall results [22]. Hadad et al. compared several deep architectures for forecasting high-frequency changes in the price ratio; by combining a Temporal Convolutional Network forecast with a classic mean-reversion rule conditioned on “prediction strength,” they sharply increased per-trade returns [12]. Reinforcement-learning agents that learn entry, exit, and stop-loss rules simultaneously have also surpassed both classical statistical approaches and earlier RL systems [13, 19]. However, most of these studies still train on a single pair chosen by traditional metrics, so overfitting remains common [10, 24].

AI techniques have also been applied directly to pair selection. Sarmento and Horta compressed price vectors using principal component analysis (PCA), applied density-based spatial clustering of applications with noise (DBSCAN) and ordering points to identify the clustering structure (OPTICS) to group similar assets, and then factored in mean-reversion criteria - while transaction costs were accounted for during the backtesting phase to evaluate each pair’s performance [34]. Chan et al. converted price series into Gramian Angular Field (GAF) images, encoded them with a convolutional autoencoder, and then used k-means clustering to find pairs expected to revert the fastest, yielding higher Sharpe ratios than earlier selection methods [4]. Xu and Luo proposed a two-level reinforcement learning framework that uses Extended Option-Critic (EOC) for dynamic trading-pair selection and Multi-Agent Deep Deterministic Policy Gradient (MADDPG) for adaptively setting opening and stop-loss thresholds and trade execution, yielding improved

returns over existing methods [44]. Even so, these studies still rely only on time-series inputs, overlooking the local visual cues in candlestick charts and other image data. Moreover, many AI-based studies rely on relatively short test periods, which limits their ability to validate long-term market performance. In contrast, we train our model on thirteen years of daily data (1991-2003) and evaluate it over roughly twenty years (2004-2024), yet still achieve consistently strong returns and Sharpe ratios, thereby demonstrating the proposed method’s robustness.

2.3 Candlesticks image data as input data

Most work that turns stock price series into candlestick images for CNN training has focused on forecasting a single asset’s price direction. Kusuma et al. created such images and compared VGG-Net, ResNet, and a generic CNN, reporting about 92% directional accuracy [23]. Hu et al. passed monthly candlestick images through a CAE to obtain latent vectors, then applied modularity-optimization clustering and a Sharpe-ratio filter to build a low-risk portfolio [15]. Hung and Chen proposed Deep Predictor for Price Movement (DPP), which takes a 20-day candlestick chart, decomposes it into overlapping 3-day sub-charts, extracts deep features via a CAE, feeds the resulting feature sequence into a Gated Recurrent Unit (GRU) - based Recurrent Neural Network (RNN) to predict the next day’s price direction, and empirically outperforms Index-Ensemble Model (IEM), Prophet, and LSTM baselines [16]. Ramadhan et al. encoded images with a GAF and fed them to a CNN-LSTM, markedly improving predictive power over a plain CNN [32]. Tuncer et al. form 2D images and use a vision transformer and a patch-based convolutional model to predict price and direction, outperforming baseline convolutional architectures [42]. No earlier study has encoded the relative movements of two assets for spread trading, so our work is the first to represent price series as images and jointly optimize pair selection and trading.

In this study, we present two key innovations. First, we transform price series into candlestick images so vision models can capture both visual patterns and the relative movement of two stocks, making full use of the data’s space-time structure. Second, we establish an integrated learning loop: each month’s realized trading performance feeds back as training data for the next month’s pair selection, allowing the model to update its rules as market conditions evolve. The combination of image-based representation and continual feedback opens a new line of research that reduces overfitting and corrects the pattern-recognition gaps noted in earlier work.

3 Image-Based Selection and Execution Framework

The pipeline forms a closed loop—image preprocessing → pair selection → trading → performance-driven adjustment of selection criteria. At the beginning of each month, raw OHLC time-series data are converted into two types of candlestick images. Each image passes through a CAE-MLP module that proposes candidate pairs and, under predefined trading rules, decides whether to open or close positions. When the trading period ends, the realized Sharpe ratio becomes the target for retraining the MLP for the following

month, allowing the pair-selection rules to adjust automatically to observed performance. Figure 1 shows the architecture of ISEPT.

3.1 Image Generation

For effective CAE training, each stock's price pattern must be preserved with minimal distortion. A 21-trading-day window is slid across the preceding 12 months, and every segment is rendered as a 64×64 px RGB candlestick image. This 21-day span captures a single monthly market rhythm [35]. We use a 21-trading-day window (≈ 1 calendar month) as a compromise between very short horizons (5 days, ≈ 1 week) that overfit transients and much longer windows (126 days, ≈ 6 months) that blur local patterns; one month preserves short-term dynamics while retaining medium-horizon context for the CAE.

Prices are log-scaled to eliminate the gap between high- and low-priced stocks. The images are then passed to the CAE, which compresses volatility and pattern information into latent vectors. The lower-left panel of Figure 1 shows a 64×64 px RGB candlestick image for Apple (AAPL) covering from 2 to 8 January 2020, illustrating how the 21-day window encodes OHLC data for the CAE to learn local price behaviour.

3.2 Pair Selection with the CAE-MLP Model

The CAE encoder maps each candlestick image into a high-dimensional latent space that retains the stock's unique volatility signature [26]. For each stock, we average all embeddings to create a single representative vector, then concatenate the two stock vectors to build a pair representation. Labels are set to the realized Sharpe ratios of the 20 best and 20 worst pairs from the previous month, and the MLP is trained as a regression model [33].

We can express these embedding and concatenation steps in mathematical form as follows. Eq. (1) defines the pair input vector $x_{ij}^{(t)}$ by concatenating the latent embeddings of assets i and j at time t . Eq. (2) specifies the average of the CAE encoder outputs over the past T months for the candlestick images $\mathbf{I}_i^{(\tau)}$. Finally, Eq. (3) shows that the next-month Sharpe Ratio prediction $\hat{S}_{ij}^{(t+1)} = f_{\text{MLP}}(x_{ij}^{(t)})$ is produced by feeding the concatenated pair vector $x_{ij}^{(t)}$ into the MLP f_{MLP} .

$$x_{ij}^{(t)} = [z_i^{(t)}, z_j^{(t)}], \quad (1)$$

$$z_i^{(t)} = \frac{1}{T} \sum_{\tau=t-T+1}^t f_{\text{CAE.enc}}(\mathbf{I}_i^{(\tau)}), \quad (2)$$

$$\hat{S}_{ij}^{(t+1)} = f_{\text{MLP}}(x_{ij}^{(t)}) = f_{\text{MLP}}([z_i^{(t)}, z_j^{(t)}]), \quad (3)$$

- $x_{ij}^{(t)}$: The input vector for the asset pair (i, j) at time t , formed by concatenating their latent embeddings.
- $z_i^{(t)}$: The average latent embedding of asset i at time t , computed over the previous T months.
- T : The number of months used to average the CAE encoder outputs; here $T = 12$.
- $\mathbf{I}_i^{(\tau)}$: The candlestick chart image of asset i at month τ .
- $f_{\text{CAE.enc}}$: The encoder function of the CAE.
- f_{MLP} : The regression function of the MLP.

- $\hat{S}_{ij}^{(t+1)}$: The predicted Sharpe Ratio for pair (i, j) in the following month $t + 1$.

This design offers two main benefits. First, because the labels come from recent trading results, the model learns to prefer truly profitable pairs rather than ones that only appeared similar in past prices. Second, each stock embedding is a visual summary with temporal order removed, so it generalizes better than fixed historical statistics when market conditions change. As a result, the MLP can quickly rank about $S^2/2$ pairwise combinations each month and select the 100 pairs with the highest expected Sharpe ratios as trading candidates.

3.3 Sharpe-Ratio-Driven Feedback Loop for Adaptive Pair Selection

A distinctive element of our framework is the reuse of Sharpe ratios from trading as labels for the next month's pair selection. After each six-month trading window, we compute the annualized Sharpe ratio for every pair, pick the top 20 and bottom 20, and feed these pairs with their ratios back into the CAE-MLP, which is retrained to forecast pairs likely to post high Sharpe ratios in the following month.

This design fulfills three aims. First, unifying the goals of selection and trading narrows the gap between the two stages. Second, training on the risk-adjusted Sharpe ratio tames the large swings that can occur when chasing raw returns alone. Third, by incorporating fresh trading outcomes each month, the model resists drift caused by shifting market conditions. The feedback loop therefore works as a self-reinforcing, performance-focused cycle in which pair selection, trading, and evaluation move continuously within a single closed loop.

4 Experiments

4.1 Dataset

We pre-process daily OHLC data for the S&P 500 from January 1990 to December 2024 and convert each series into images for later analysis. At every pair-selection date, we keep only the constituents whose closing-price record has no gaps in the twelve months before and six months after that date. For each stock that passes this filter, we join its most recent twelve months of prices and slide a 21-trading-day window forward one day at a time, rendering each segment as a 64×64 px candlestick chart. The resulting images form the input to CAE training. The set of single-asset images is randomly split 70% for training and 30% for validation [40]. We train for 13 years, longer than the three to five years and occasionally ten years common in prior work, to capture a broad range of market regimes, and we evaluate out of sample for about 20 years to test robustness and generalization across diverse conditions.

4.2 Model Architecture and Training

4.2.1 CAE Model. The CAE receives 64×64 px RGB images and follows a three-stage, symmetrical encoder-decoder design [26]. The encoder applies $64 \rightarrow 128 \rightarrow 256$ convolutional channels and three 2×2 max-pooling layers, reducing the spatial resolution from $64 \rightarrow 32 \rightarrow 16 \rightarrow 8$ to produce an $8 \times 8 \times 256$ latent tensor. The decoder mirrors these steps with up-sampling and convolutions ($256 \rightarrow 128$

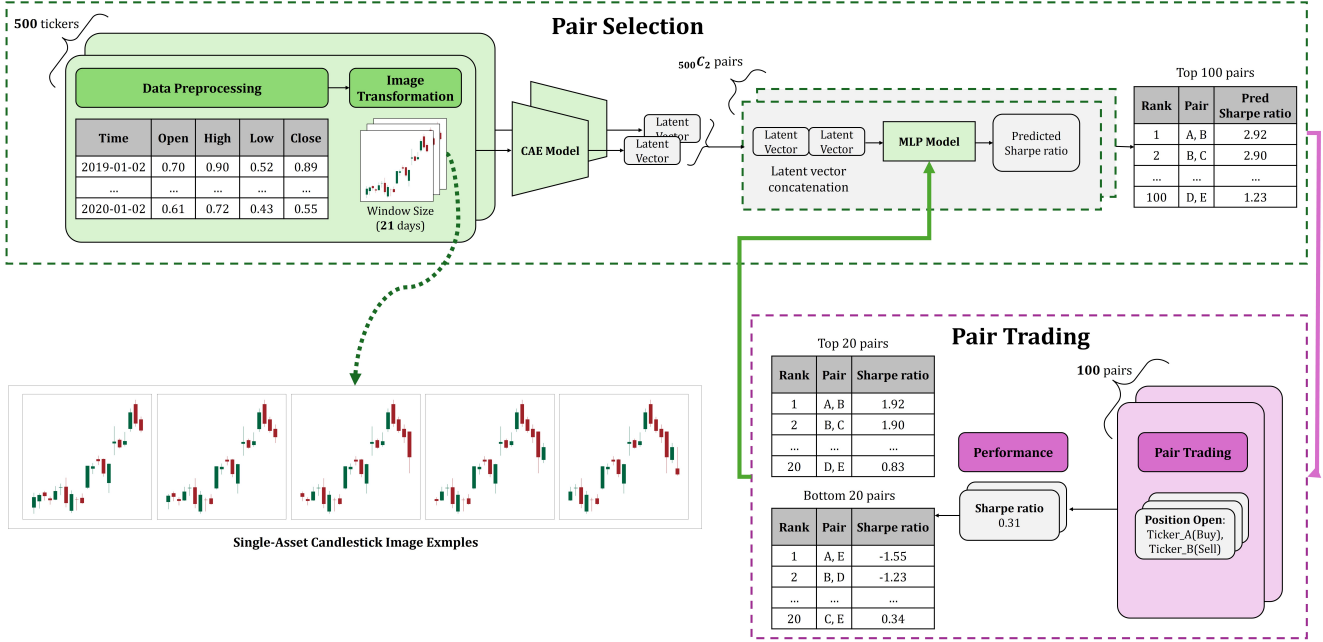


Figure 1: The framework of ISEPT. The diagram outlines the full pipeline - converting OHLC time-series into candlestick images, extracting stock-level representations with a CAE, and feeding these into downstream models for pair selection and trading. The bottom-left image provides a concrete example: a 64×64 px RGB candlestick chart of Apple (AAPL) spanning 2-8 January 2020. Within its 21-trading-day window, colour and candlestick length encode OHLC values, vividly capturing the local price patterns and volatility that the CAE is designed to learn.

→ 64 → 3 channels), restoring the original resolution, and ends with a sigmoid layer to confine pixel values to $[0, 1]$ [33]. Batch Normalization and PReLU follow every convolution to enhance training stability and expressive power [14, 17].

Training uses all ticker images from the 12 months immediately preceding each pair-selection date, split 70%, 30% into training and validation sets. We minimize mean-squared error (MSE) for up to 20 epochs with an initial learning rate of 1×10^{-4} . The learning rate is halved every five epochs, and training stops early if validation loss worsens in three consecutive checks [30]. The model converges rapidly and reconstructs images that are visually indistinguishable from the originals, ensuring that high-level price features are preserved for subsequent stages.

4.2.2 MLP Model. For each stock, we average its latent vectors (16,384 dimensions) to obtain one representative vector, then concatenate the two stock vectors to form a 32,768-dimensional pair representation. Principal-component analysis reduces this to 512 dimensions, which serve as the MLP input.

The MLP is a fully connected network with the layout LayerNorm-1024-512-128-1 [33]. Each hidden layer employs ReLU activation and 0.5 dropout to learn complex nonlinear relations while curbing over-fitting [28, 39]. The final node is linear and outputs the predicted Sharpe ratio for the pair. Thus, the model regresses Sharpe ratios directly from latent price patterns.

Training uses a batch size of 512, an initial learning rate of 1×10^{-3} , the Adam optimizer, and MSE loss; early stopping is triggered if validation loss fails to improve for five straight epochs [20]. The trained MLP ranks all combinations each month and selects the 100 pairs with the highest expected Sharpe ratios for trading.

4.3 Baselines

We benchmark our approach by splitting the workflow into a separate pair-selection stage and a trading stage, following Gatev et al. (hereafter GATEV) [11] and Vidyamurthy (hereafter VIDYAMURTHY) [43].

In GATEV method, cumulative total returns for all stocks over a 12-month formation window are first normalized to 1. The daily squared price differences between every pair are then summed, and the 100 pairs with the smallest sums-i.e., the most similar historical price paths-are retained. This minimum-distance rule remains one of the most widely used criteria in classical statistical arbitrage. Meanwhile, VIDYAMURTHY selection procedure begins by filtering the universe to pairs whose absolute return correlation is high. It then applies the Engle-Granger cointegration test to those candidates, keeping only pairs whose regression residuals are stationary. The combination of correlation pre-screening and cointegration testing efficiently extracts pairs whose long-run equilibrium relationship is statistically verified.

For trading, GATEV employs a 2σ mean-reversion rule. Each selected pair is monitored for six months. When the spread-the price difference between the two stocks-exceeds twice its formation-period standard deviation, USD 1 of the rich stock is shorted and USD 1 of the cheap stock is bought, creating a dollar-neutral long-short position. The position is closed when the spread re-crosses its mean; if convergence does not occur by the end of the trading window, any remaining positions are liquidated. Meanwhile, in VIDYAMURTHY trading regime, an optimal threshold $\Delta \approx 0.75\sigma$ is derived from the spread's empirical distribution. A dollar-neutral position is opened when the spread departs by $\pm\Delta$ from the mean and is closed immediately upon mean reversion to zero. Both baselines forbid overlapping positions in the same pair.

5 Results

This section evaluates five pair-trading strategies using daily S&P 500 OHLC data from January 2004 through June 2024. Results are reported for (i) the full sample and (ii) two stress windows in which market shocks were concentrated-the Global Financial Crisis (December 2007-June 2009) and the COVID-19 pandemic (February 2020-April 2020).

The strategies are configured as follows. Our model (hereafter ISEPT) uses the CAE-MLP pair-selection module with a Sharpe-ratio feedback loop that updates model parameters on a rolling basis. Strategy ISEPT + GATEV combines the same CAE-MLP selection with Gatev's classical trading rule: a position is entered when the spread exceeds $\pm 2\sigma$ and closed when it reverts to $\pm 1\sigma$. Strategy ISEPT + VIDYAMURTHY also uses CAE-MLP selection but applies Vidyamurthy's rule, opening a position on the day after the spread moves beyond $\pm 0.75\sigma$ and closing it once the opposite $\pm 0.75\sigma$ band is crossed. The two purely statistical baselines, GATEV and VIDYAMURTHY, follow their original selection and trading protocols. All strategies incorporate a one-basis-point one-way transaction cost when computing performance metrics [3].

Section 5.1.1 reports the performance of the five strategies for the full 2004 – 2024 sample, and Section 5.1.2 examines their behaviour during the Global Financial Crisis and the COVID-19 pandemic.

5.1 Performance Over the Entire Period

Table 1 summarizes the key statistics for the full sample, including trading frequency, profitability, and risk-adjusted returns, for the four competing strategies over 2004–2024. Their month-by-month trajectories are illustrated by the performance curves shown in Figure 2.

Over the full sample, trading frequency differed markedly across the four strategies. Count, which records how often positions are initiated and thus reflects trading intensity, averaged 3.50 trades per month for ISEPT + VIDYAMURTHY, the most active of the set; ISEPT + GATEV followed at 1.28, while the classical Vidyamurthy and GATEV baselines posted 2.83 and 1.55, respectively. The higher Counts for the ISEPT variants show that our pair-selection module generates more signals yet still improves overall performance, suggesting it identifies opportunities more efficiently.

Profitability was likewise superior under our framework. Annualized return on investment(ROI), which expresses cumulative

profit on a yearly basis and makes results comparable across horizons, reached 18.97% for ISEPT + GATEV and 16.48% for ISEPT + VIDYAMURTHY, far surpassing the 3.52% and 4.02% delivered by the GATEV and Vidyamurthy baselines. Risk-adjusted measures tell a consistent story: the Sharpe ratio - excess return per unit of total volatility [37] - rose from 0.4001 to 0.6837 when GATEV's trading rules were paired with ISEPT, and from 0.4724 to 0.7892 when Vidyamurthy's rules were augmented in the same way. The Sortino ratio, which focuses solely on downside volatility and therefore rewards strategies that avoid large losses [38], improved in parallel, climbing from 0.8935 to 1.3102 and from 0.8128 to 1.3419.

Absolute risk levels, as captured by maximum drawdown(MDD) and volatility, were somewhat higher for the ISEPT-based approaches, with MDD in the low-ten-percent range and volatility around 0.19 [25]. Even so, the Calmar ratio - annual return divided by MDD, showing how efficiently profit is generated relative to severe losses [45] - reached 3.1572 and 2.2398, exceeding the 2.6675 and 1.3306 posted by the traditional methods; the greater increase in reward more than compensated for the additional risk.

Finally, the Hit Ratio, which measures the fraction of trading days that end in a positive realized return and distinguishes steady gain-builders from strategies reliant on a few large wins [41], stood at 50.36% for ISEPT + GATEV and 48.70% for ISEPT + VIDYAMURTHY, both higher than their statistical counterparts. These advantages persist even after subtracting a one-basis-point one-way transaction cost, confirming that the ISEPT module dynamically selects high-quality pairs and improves Count and Hit Ratio in tandem, lifting overall performance under identical trading rules.

5.2 Performance During Crisis Periods

This section compares the four strategies during two episodes of severe market stress in the United States: the Global Financial Crisis (December 2007-June 2009) and the initial plunge triggered by the COVID-19 pandemic (February 2020-April 2020). Table 2 summarizes the headline performance statistics for the Global Financial Crisis, while Figure 3 plots the contemporaneous trajectories of each metric. Similarly, Table 3 reports the corresponding results for the early COVID-19 plunge, and Figure 4 visualizes their evolution over that turbulent period.

During the Global Financial Crisis, ISEPT + GATEV executed an average of just 1.66 trades per month, yet delivered the highest annualized return on investment (ROI=51.63%) and posted the best risk-adjusted figures - Sharpe ratio 1.2405, Sortino ratio 2.1886, and Calmar ratio 5.2139. Although its MDD and volatility increased, the gains more than compensated for the added risk. ISEPT + VIDYAMURTHY traded most actively (4.96 trades per month) and still achieved an ROI of 33.96%, a Sharpe ratio of 1.0944, and a Sortino ratio of 1.8201, comfortably outperforming the classical Vidyamurthy baseline (ROI 10.31%, Sharpe 0.6003). By contrast, the Gatev and Vidyamurthy benchmarks produced single-digit ROIs and Sharpe ratios in the 0.5-0.6 range.

The advantages persisted in the early COVID-19 crash. ISEPT + GATEV again led the field with an ROI of 45.45%, a Sharpe ratio of 0.9001, and a Calmar ratio of 5.1712. Maintaining a high trading frequency of 6.53 trades per month, ISEPT + VIDYAMURTHY recorded an ROI of 24.33% and a Sharpe ratio of 0.8038, clearly beating the

Table 1: Performance Over the Entire Period

Trading Type	Count	ROI	Sharpe Ratio	Sortino Ratio	Maximum Drawdown	Calmar Ratio	Volatility	Hit Ratio
GATEV	1.5528	0.0352	0.4001	0.8935	0.0455	2.6675	0.0808	0.4923
VIDYAMURTHY	2.8255	0.0402	0.4724	0.8128	0.0495	1.3306	0.0781	0.4804
ISEPT + GATEV	1.2776	0.1897	0.6837	1.3102	0.1045	3.1572	0.1987	0.5036
ISEPT + VIDYAMURTHY	3.4984	0.1648	0.7892	1.3419	0.1102	2.2398	0.1912	0.4870

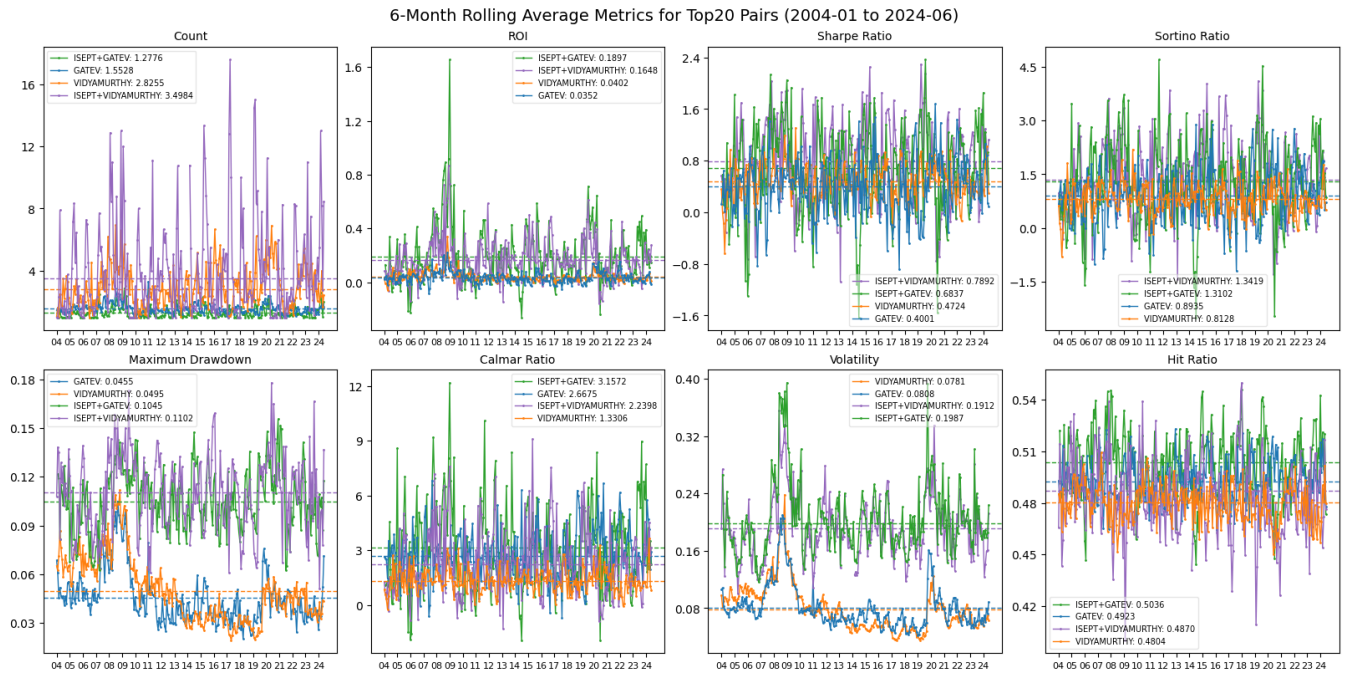


Figure 2: Performance Over the Entire Period. The figure plots the evolution of each performance metric for all strategies from January 2004 to June 2024.

Table 2: Performance During the Global Financial Crisis

Trading Type	Count	ROI	Sharpe Ratio	Sortino Ratio	Maximum Drawdown	Calmar Ratio	Volatility	Hit Ratio
GATEV	2.0368	0.0955	0.5036	0.9266	0.0833	2.4812	0.1535	0.4909
VIDYAMURTHY	4.2760	0.1031	0.6003	0.9746	0.0910	1.4838	0.1548	0.4718
ISEPT + GATEV	1.6642	0.5163	1.2405	2.1886	0.1214	5.2139	0.2874	0.5144
ISEPT + VIDYAMURTHY	4.9602	0.3396	1.0944	1.8201	0.1291	3.4115	0.2541	0.4839

Table 3: Performance During the COVID-19 Pandemic

Trading Type	Count	ROI	Sharpe Ratio	Sortino Ratio	Maximum Drawdown	Calmar Ratio	Volatility	Hit Ratio
GATEV	2.2167	0.0998	0.7871	1.5178	0.0569	3.7232	0.1179	0.4985
VIDYAMURTHY	4.4348	0.0907	0.6642	0.9962	0.0534	1.5989	0.1112	0.4726
ISEPT + GATEV	1.9750	0.4545	0.9001	1.6114	0.1380	5.1712	0.2832	0.4984
ISEPT + VIDYAMURTHY	6.5294	0.2433	0.8038	1.2511	0.1374	1.7157	0.2557	0.4775

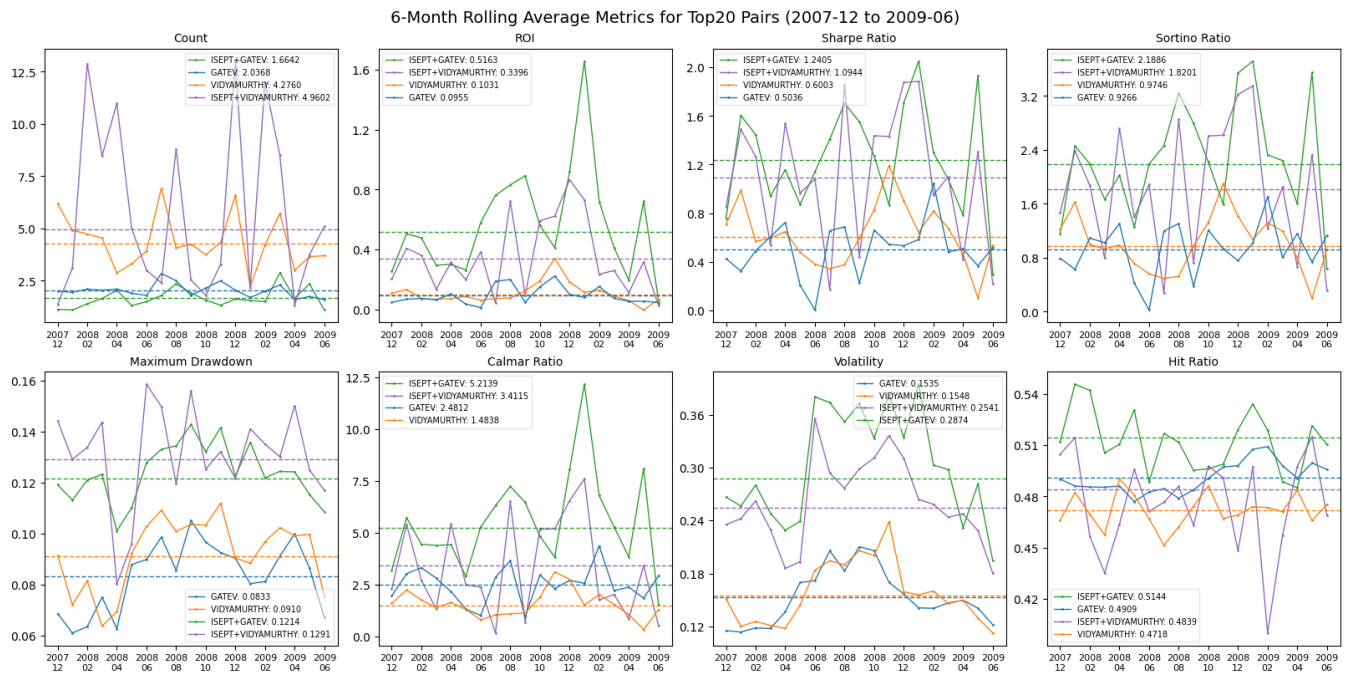


Figure 3: Performance During the Global Financial Crisis. The figure traces the trajectory of each performance metric for all strategies during the Global Financial Crisis.

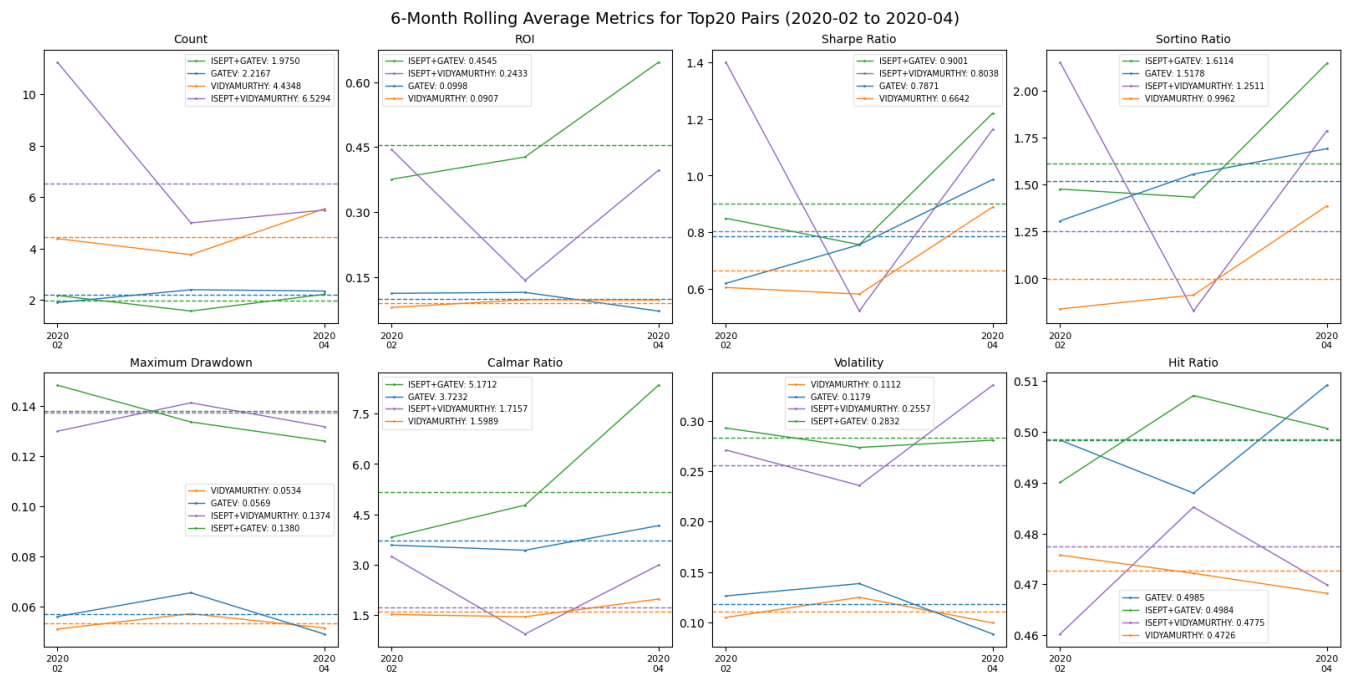


Figure 4: Performance During the COVID-19 Pandemic. The figure plots the trajectory of each performance metric for all strategies during the first three months of the COVID-19 crash.

traditional Vidyamurthy rule (ROI 9.07%, Sharpe 0.6642). In both hybrid strategies, the Hit Ratio remained near 48–50%, indicating that entry signals were more accurate than those of the statistical baselines even amid extreme volatility.

During crises, market price adjustments often slow, leaving stock prices well above or below their intrinsic values. In such periods the ISEPT strategy tends to outperform, implying that its gains arise from exploiting these temporary mispricings [2, 27].

Across the full sample and in both crisis windows, the proposed framework (ISEPT) delivers stronger results than the statistical baselines on every key measure including ROI, Sharpe, Sortino, and Calmar ratios. Two design features drive this edge. First, converting prices to candlestick images lets the model detect inter-asset links that raw time-series values miss. Second, a closed loop sends each month's realized Sharpe ratio back into training for the next month's pair selection, so the MLP constantly relearns which pairs have been profitable under current conditions.

Because the MLP is trained on pairs that have just performed well, it adjusts quickly when the covariance structure shifts during crises, extracting the highest return for a given level of risk. All metrics are reported net of a one-basis-point transaction cost per trade, confirming that the image-based approach continues to beat traditional statistical methods even after trading expenses.

6 Conclusion

We introduce ISEPT, an end-to-end pair-trading framework combining candlestick-image inputs with a Sharpe-ratio feedback loop can earn steady risk-adjusted returns even during severe market swings. Using S&P 500 data from January 2004 to June 2024, the proposed model beat standard statistical methods and held positive Sharpe ratios alongside solid profits through both the Global Financial Crisis and the COVID-19 period. The gains come from (i) recasting time-series data as two-dimensional images so a CAE can capture local price patterns and (ii) linking pair selection, trading, and evaluation in a loop that updates criteria every month.

Several limits remain. We fix execution to isolate selection effects and ensure fair comparability; the framework is execution-agnostic, and joint selection-execution optimization is left for future work. Image generation and the large number of pair evaluations require heavy computation. Tests cover only S&P 500 stocks, and transaction costs and liquidity are simplified. Future research should extend the framework to multiple asset classes and cross-market settings, widening the scope for image-based, integrated pair trading.

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